



DEEP LEARNING TAKEAWAYS



Chapter:

Transformers

Word Embeddings

- 1** Word embedding is a way to represent a word in a numeric format such that it captures the semantic meaning of that word
- 2** Word2vec, Glove etc are popular techniques to produce static word embeddings
- 3** With word embeddings, you can do math like $\text{King} - \text{man} + \text{woman} = \text{Queen}$ or $\text{Russia} - \text{Moscow} + \text{Delhi} = \text{India}$

Overview of Encoder and Decoder

- 1** Transformer architecture has two parts (1) Encoder (2) Decoder
- 2** The purpose of encoder is to produce a contextual embeddings for each word (more precisely a token) in a given input sentence
- 3** The purpose of decoder is to produce an output sequence which can be a word (for the next word prediction task) or a sequence (such as translated sentence in case of language translation)
- 4** BERT and GPT are examples of specific models that are based on a transformer architecture

Attention Mechanism

- 1** Attention mechanisms allow Transformers to focus on relevant parts of the input sequence for each output, improving context understanding.
- 2** Self-attention computes the relationships between all input elements, capturing dependencies regardless of their position.
- 3** Key components of attention include queries, keys, and values, which determine how much focus is given to different parts of the input.

Multi Headed Attention

- 1** Multi-headed attention enables Transformers to capture diverse relationships in the data by learning multiple attention patterns simultaneously.
- 2** Each attention head computes self-attention independently, focusing on different parts of the input sequence.
- 3** Outputs from all heads are concatenated and transformed to create a richer representation of the input.
- 4** Multi-headed attention improves the model's ability to understand complex patterns and long-range dependencies.
- 5** It is a key component in Transformer Architecture.

Decoder

- 1** The decoder in Transformer architecture generates the output sequence step-by-step, one token at a time.
- 2** It uses masked self-attention to ensure predictions depend only on previously generated tokens.
- 3** The decoder integrates encoder outputs through cross-attention to incorporate contextual information from the input sequence.
- 4** Fully connected layers in the decoder refine the processed information for final token prediction.
- 5** The decoder is central to tasks like language translation and text generation, where sequential output is crucial.

How Transformers are Trained?

- 1** In Self-supervised learning, labels are generated from the data itself without requiring manual annotations.
- 2** Casual Language Modeling (CLM) and Mask Language Modeling (MLM) are self-supervised learning approaches used to train transformers
- 3** GPT uses CLM whereas BERT uses MLM